CARLOS PEREZ MENDOZA

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Professional summary

AI/Machine Learning Scientist with a Ph.D. in Mathematics & Statistics and 7+ years of experience in machine learning, reinforcement learning, and probabilistic modeling. Skilled in deep learning architectures with a proven ability to design scalable, high-performance AI solutions. Experienced in bridging research rigor with industry applications, delivering impactful outcomes in finance, risk management, and digital health.

SKILLS & EXPERTISE

Domains of Application: Finance, Risk Management, Digital Health.

- Deep Learning (PyTorch, TensorFlow)
- Reinforcement Learning
- Bayesian Optimization
- Probabilistic Modelling
- Statistical Modelling
- Quantitative Analysis
- Data Management (Python,SQL,R)
- Data Visualization
- Signal Processing

SELECT ACHIEVEMENTS

Published multiple peer-reviewed papers on deep learning, reinforcement learning, and AI models, with applications in algorithmic trading, risk management, and predictive modeling.

Professional Experience

Scientifique de recherche appliquée — Montreal, Canada
JACOBB - Centre d'intelligence artificielle appliquée, Direction Recherche & Innovation — Mar 2023-Jul 2024

- Designed and implemented AI-driven pipelines for startup-investor matchmaking, startup recommendation, and wearable-based diabetes monitoring, leveraging NLP, web scraping, and deep learning (transformers), achieving accuracy rates of 83%, 85% and 87%, respectively.

Quantitative Research Analyst — Montreal, Canada
Plant-E Corp, Algorithmic Trading Department — Jul 2022-Nov 2022

- Developed and implemented machine learning-based methodologies for quantitative trading and electricity price forecasting in North American Power Markets, leveraging reinforcement learning to improve day-ahead and futures price forecasting precision by 5% while meeting regulatory requirements.

Presidential Advisor — Mexico City
Private Assistance Board of the Government of Mexico City — Oct 2020-Ago 2021

- Applied statistical modeling and optimization techniques to evaluate institutional performance, design data-driven solutions, and improve operational efficiency for long-term sustainability.

Head of the Office of Quantitative Health Plan Studies — Mexico City
Bank of Mexico (Central Bank of Mexico), Technical Sub-Management of the Health Plan — Nov 2017-Jan 2019

- Developed and deployed probabilistic models, automated data processes, and implemented control systems to optimize resource usage and improve operational accuracy.

Market Research Analyst — Mexico City
Bank of Mexico (Central Bank of Mexico), Health Plan Evaluation Sub-Management — Mar 2015-Oct 2017

- Built anomaly detection and predictive models to prevent fraud, optimize supplier performance, and achieve measurable cost savings.

Education

Ph.D. Mathematics & Statistics | Concordia University, Canada (GPA: 4.3/4.3) | Sep 2021–Apr 2025
M.Sc. Mathematics, Mathematical Statistics & Probability | UNAM, Mexico (GPA: 9.9/10) | Jan 2019–Jul 2020
M.Sc. Modern Applications of Mathematics | University of Bath, UK | Jan 2020–Jul 2020
Specialist Degree in Financial Engineering | UNAM, Mexico (GPA: 9.6/10) | Aug 2021–Jul 2022
B.Sc. Actuarial Sciences – UNAM, Mexico (GPA: 9.20/10) | Aug 2010–Dec 2014
Graduate Diploma in Statistical Techniques & Data Mining | UNAM, Mexico (GPA: 10/10) | Jan 2016–Oct 2016

Certification & Training

Artificial Intelligence: An Overview Specialization (5 Courses) – Politecnico di Milano, 2023
Introduction to Data Science in Python – University of Michigan, 2020
Introduction to TensorFlow for AI, ML, and DL – DeepLearning.AI, 2019
Deep Learning Specialization (5 Courses) – DeepLearning.AI, 2018
Quantum Computing 101 – IBM, 2018
Microsoft Excel and Visual Basic for Applications (VBA)

Languages

English: Full Professional **French:** Intermediate **Spanish:** Native